

WORK EXPERIENCE

ML Engineer, Founding Team (Intern) <i>Veach & Company</i>	Jul 2025 – Dec 2025 <i>New York, USA</i>
- Built and deployed a production retrieval and ranking pipeline in Python (ingestion, embedding, candidate retrieval, LLM-assisted re-ranking) over a manually sourced corpus of ~100 investor profiles, lifting top-5 relevance ~40% over the keyword baseline. - Owned model evaluation: defined quality metrics with product and operations stakeholders, curated and labeled a 100-example ground-truth set, and gated every prompt, model, and ranking change on it before release. - Benchmarked GPT-4-class and open-source models on relevance, structured-output reliability, latency, and cost; shipped tiered routing with embedding caching and batched inference, cutting serving cost ~30% at constant quality.	
Software Engineer, Machine Learning <i>Accenture</i>	Jun 2022 – Oct 2023 <i>Pune, India</i>
- Compared Logistic Regression, XGBoost, and anomaly-detection models on 1M+ daily banking transactions using cross-validation and AUROC, selecting the model and decision threshold that reached AUROC 0.91 and cut false positives 20%, with feature engineering and feature selection. - Operated the models in production: tracked per-run AUROC and precision/recall across versioned splits, traced degradation to upstream data and feature changes, and routed low-confidence predictions to human review. - Designed and analyzed A/B tests on the payments page and login journey across a rollout of 8 features (processing time –15%), tracking Android/iOS response times across service providers; technical point of contact for the client, running weekly demos.	
Associate Data Engineer <i>Accenture</i>	Dec 2020 – May 2022 <i>Pune, India</i>
- Built high-volume PySpark and Databricks pipelines unifying core banking, ISP, and transaction-log data into analysis-ready Snowflake tables for model training (data preparation –25% time); optimized SQL (window functions, CTEs, stored procedures, indexing) powering 500+ KPIs, cutting runtime 35%. - Integrated third-party systems over REST APIs (OAuth, API keys, webhooks) for the payments feature, built the two-factor-authentication notification flow on Azure Notification Hubs, and supported model deployment on AWS.	
Graduate Research Assistant (CUBIT) <i>Stony Brook University</i>	Apr 2026 – Present <i>Stony Brook, USA</i>
Automated data-quality rules over sensor streams from 6 device types (40 participants each), encoding validity and completeness thresholds that flag device failures before analysis and cutting per-participant processing from 10–15 minutes to ~5.	

PROJECTS

CTR Prediction, Deep Ranking Model vs. Gradient Boosting PyTorch, XGBoost, Criteo display-ads data
Training a deep CTR model with embedding tables for 26 high-cardinality features against an XGBoost baseline on AUC and log loss; evaluating calibration alongside ranking quality, with negative downsampling and prior correction for severe class imbalance.
Kathavani, Grounded RAG (NLP) Python, pgvector, HNSW, BGE embeddings & reranker GitHub
Built two-stage retrieval over 11,632 chunks (cosine top-100, reranker to top-12) with an automated entity-grounding check scoring every output against retrieved context to catch hallucinations.
Multimodal Compliance Detection (Vision + NLP) Azure OpenAI, Azure AI Search, LangGraph GitHub
Built a pipeline extracting OCR and transcripts from video, retrieving policy passages cross-modally, and producing structured risk scores; low-confidence outputs escalate to human review.

LEADERSHIP & AWARDS

- Treasurer, Graduate Student Organization, Stony Brook University (May 2025 – May 2026): administered a \$900K budget across ~12 funding lines, reviewed 1,300+ funding applications, and presented to the Provost and University Senate. - Accenture Celebrates Excellence Award (STAR) for an internal tool that cut a 20-minute Kibana lookup workflow to 2 minutes; Accenture Merit Holder FY2021.

EDUCATION

Stony Brook University <i>M.S. Applied Mathematics and Statistics, GPA 3.7/4.0 Machine Learning, Statistical Learning, NLP</i>	Aug 2024 – Dec 2026 <i>Stony Brook, New York</i>
Savitribai Phule Pune University <i>Bachelor of Engineering in Computer Engineering, GPA 3.86/4.0</i>	May 2016 – May 2020 <i>Pune, India</i>

TECHNICAL SKILLS

Languages: Python, SQL, C++, R, TypeScript
ML Frameworks & Methods: PyTorch, TensorFlow, Scikit-learn, XGBoost; deep learning, ranking and recommendation, NLP/RAG, anomaly detection, feature engineering and selection, A/B testing, calibration, model monitoring
Data & Serving Infrastructure: Apache Spark (PySpark), Databricks, Airflow, dbt, Snowflake, PostgreSQL, Docker, AWS (EC2, S3, SageMaker), Azure
LLM Systems: LLM APIs (OpenAI, Azure OpenAI, Anthropic), embeddings, rerankers, vector search (pgvector), LangGraph agents